

Govind Kumar

Machine Learning Engineer

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Summary

Machine Learning Engineer building real-time computer vision for Intelligent Transportation Systems — vehicle detection and classification, incident detection, number-plate recognition — on roadside edge hardware without GPUs. Optimise models with OpenVINO and TensorRT to hold accuracy inside fixed frame budgets. Peer-reviewed publication in mobile inference optimisation.

Technical Skills

Languages: Python, C++, SQL, TypeScript, Kotlin

ML & Computer Vision: PyTorch, TensorFlow, TensorFlow Lite, OpenCV, YOLOv8/v11, SSD MobileNet, scikit-learn, NumPy, Pandas

Optimisation & Serving: OpenVINO, TensorRT, post-training quantisation, structured pruning, ONNX/IR export, FastAPI, Flask, REST APIs

Infrastructure & Data: Docker, Git, GitHub Actions CI, Linux, PostgreSQL, MySQL, SQLite, ChromaDB, LangGraph, asyncio, multi-threaded RTSP/UDP ingestion, pytest

Experience

AI/ML Developer

Feb 2026 – Present

Metro Infrasyys Private Limited, Noida

- Architected a two-stage ATCC inference pipeline separating per-camera vehicle localisation from a shared classification service, raising multi-camera throughput by reusing one classifier across streams.
- Deployed Video Incident Detection (VIDS) to edge hardware with **OpenVINO**, mapping stalled-vehicle detections to discrete stop-event triggers to suppress false alarms on slow-moving traffic.
- Sustained **99.9% stream uptime** in production via multi-threaded RTSP ingestion with automated network probing and reconnection, and served model outputs to traffic dashboards over **REST APIs**.
- Built UDP radar monitoring services with software filters that reject hardware-induced anomalies at ingest.

Software Developer Trainee

Jul 2025 – Jan 2026

Metro Infrasyys Private Limited, Noida

- Reached **90.8% classification accuracy at 25 FPS** on YOLO-based detectors through post-training quantisation and structured pruning.
- Engineered three production Python systems — **ANPR**, VIDS with vehicle stop detection, and Automatic Vehicle Classification for automated tolling — and decomposed monolithic codebases into modular services to shorten deployment cycles.
- Deployed and configured VASD and ANPR installations on-site across four-lane highway sites (Rithwik, KNR-Kerala), resolving gateway connectivity and camera alignment faults.

Database Engineering Intern

Aug 2023

EPlanet Technologies, New Delhi

- Cut query execution time **22%** on high-traffic **MySQL** datasets through index design and schema refinement.

Projects

pipeline-sentry — static analysis and LLM triage for ML pipelines

Python, LangGraph, Kuzu, FastAPI

- Maps a Python repository into an embedded Cypher property graph and an AST-aligned vector index at **913 files/s** across 8 processes (3.6× sequential), feeding hybrid retrieval to a **LangGraph** agent workflow served over **FastAPI**.
- Six ML/CV rules general SAST tools miss — unanchored model paths, unreleased camera handles, unsafe `torch.load` — validated on a two-sided fixture: all 11 labelled defects located, **zero findings on the clean counterpart**, gated in CI. Built the eval harness measuring LLM triage against that baseline and published the negative result: an 8B local model degraded output.

Real-Time Mobile Object Detection

TensorFlow Lite, Kotlin, Android

- Quantised SSD MobileNet detector reaching **85% accuracy at 30 FPS** for CPU-only on-device inference with no GPU delegate. Implementation behind the publication below.

FRIDAY — local-first voice agent

Python, Ollama, OpenVINO

- ReAct loop over a local Llama 3.1 chaining tools autonomously; filesystem-discovered plugins, on-device speech recognition and synthesis, no cloud inference.

Publication

Mobile Real-Time Object Detection with SSD MobileNet

2025

Int. Journal of Scientific Research in Engineering and Management (IJSREM) · DOI: [10.55041/IJSREM47895](https://doi.org/10.55041/IJSREM47895)

Education & Certifications

B.Tech, Artificial Intelligence & Data Science

2021 – 2025

University School of Automation & Robotics, GGSIPU Delhi · GPA: **8.25/10**

Full Stack Developer (GeeksforGeeks) · Data Analytics & Visualization (Forage) · Power BI (Skillup)